$$
\begin{aligned}
\text{Maximize} \quad & c_1x_1 + c_2x_2 + c_3x_3 + c_4x_4 \\
& \quad + c_5x_5 + ... + c_mx_m && (1)\\
\text{s.t.} \quad & a_{i1}x_1 + a_{i2}x_2 + a_{i3}x_3 \\
& \quad + a_{i4}x_4 + ... + a_{im}x_nm \le b_i \qquad \forall\ i \in [\![n]\!] && (2)\\
& \mathbf{x} \ge \mathbf{0} && (3)
\end{aligned}
$$

$$
\begin{aligned}
\text{maximize} \quad & \mathbf{c}^\top \mathbf{x} && (4)\\
\text{s.t.} \quad & \mathbf{a}_i^\top \mathbf{k} \le \mathbf{b} \qquad \forall i \in I && (5)
\end{aligned}
$$

$$
\begin{aligned}
& d_1d_1 + d_2x_2 \\
& \quad + d_3x_3 + d_4x_4 \le f && (6)\\
& \begin{bmatrix} 1 & 0 & & 0 \\ 0 & 1 & & 0 \\ & & & \\ 0 & 0 & & 1 \end{bmatrix} \mathbf{x} = \mathbf{x} whomadethisonesolong && (7)\\
& \mathbf{x} \ge \mathbf{0} && (8)
\end{aligned}
$$